



vehicles, drones, defence systems, robotics, warehouse automation, and smart infrastructure. It resists sunlight and interference from other lidars, and



it avoids errors such as blooming and ghosting around reflective objects. The sensor provides 360-degree horizontal and 90-degree vertical coverage for near-field detection, object classification, obstacle avoidance, and situational awareness. Omni has a range of 80 metres in an 85mm diameter package. Combining coverage with per-point velocity data improves the detection of moving objects, path planning, and decision-making. It works in rain, dust, fog, and high-pressure wash-down environments.

Aeva
<https://www.aeva.com>

Indian AI laser system on unmanned ground vehicle

An Indian startup, olee.space, has unveiled a 2-kilowatt AI laser system on an unmanned ground vehicle



(UGV). The system can neutralise, dazzle, or damage personnel, vehicles, and equipment within seconds. The UGV measures 1.52m x 0.91m (5ft x 3ft), weighs under 250kg, and runs on a diesel engine. It has a range of over 130km, extendable with extra fuel,

supporting missions in remote areas. Its size allows deployment in urban zones, borders, forests, deserts, and rough terrain. The system delivers laser engagement and has AI and machine-learning algorithms for navigation, target recognition, and engagement. It can operate at the push of a button, reducing risks to personnel.

olee.space
<https://www.samsung.com/in>

200MP sensor for zoom cameras

Samsung has released the 200MP ISOCELL HP5 camera sensor. It has 0.5µm pixels, which work better for zoom and periscope cameras than for



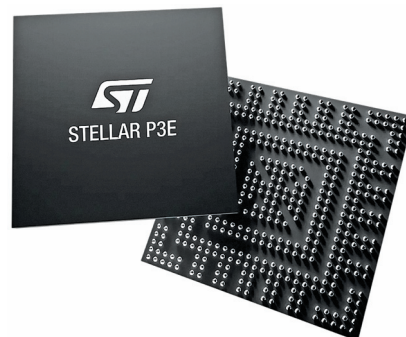
main cameras. The sensor is 2.5/3.9cm (1/1.56-inch), smaller than older sensors. To capture light and reduce noise, it uses front deep trench isolation, dual vertical transfer gate, and DTI Center Cut (DCC) structures. microlens, anti-reflection layer, and deep trench isolation improve colour and low-light performance. Made for Zoom cameras, it can also support main cameras on mid-range phones. It provides 2x in-sensor zoom and up to 6x lossless zoom. Tetrapixel technology combines pixels in low light to create 12.5MP or 50MP images.

Samsung
<https://www.overt.com>

Automotive microcontroller

STMicroelectronics announced the Stellar P3E, its first automotive microcontroller with AI acceleration for edge vehicle intelligence. It targets software-defined vehicles and supports multi-function X-in-1 electronic control

units to reduce system cost, weight, and complexity. The Stellar P3E has a neural processing unit that enables



real-time AI processing for automotive applications. This allows virtual sensors and sensing, with inference processing up to 30x more efficient than standard MCU cores. Applications include predictive maintenance, EV charging, and deployment of vehicle features with fewer sensors and modules.

STMicroelectronics
<https://www.st.com/content>

Galvanically isolated current sensor

Allegro MicroSystems Inc. has introduced the ACS37200, a galvanically isolated current sensor for high-current systems that reduces power loss and



enables smaller, more efficient designs. In applications such as hybrid/electric vehicles, industrial automation, AI data centres, and solar inverters, traditional shunt resistors generate heat and waste energy, thereby limiting efficiency. The ACS37200 cuts energy loss by 90%, freeing more power for the load and improving vehicle range or lowering data centre costs. It reduces board space by 95%, while heat and integrated isolation eliminate the need for heatsinks and external components.

Allegro MicroSystems Inc.
<https://www.allegromicro.com>